

# LAB: Medical Image Analysis with Federated Learning

Yuriy Kochura, Ivan Zhuk and Yuri Gordienko

*“Efficiency is doing the thing right. Effectiveness is doing the right thing.”*

– Peter Drucker

## Introduction

In this practical session, we'll explore the application of **Federated Learning (FL)** in healthcare through medical image analysis using the **MedMNIST** dataset collection. We examine how FL enables collaborative model training while preserving data privacy and security — a critical requirement across several sensitive domains. These include medical image analysis, where models are trained collaboratively across multiple participants with support for both 2D classification and 3D segmentation tasks; drug discovery, where FL facilitates collaborative molecular property prediction; and medical devices, where distributed data analytics support AI applications deployed on edge systems.

Key challenges in healthcare federated learning arise from the need to develop collaborative AI systems under strict privacy regulations and heterogeneous operational conditions. Healthcare and life sciences are governed by stringent regulatory requirements, such as HIPAA and GDPR, which impose strong limitations on the use and sharing of sensitive patient data. In addition, data heterogeneity is a major issue due to differences in imaging equipment and acquisition protocols, variations in patient demographics, inconsistencies in data formats and clinical standards across institutions. Finally, federated model training introduces challenges related to model performance and evaluation, including the design of robust training schemes for domain-specific tasks and ensuring reliable assessment of clinical accuracy across distributed and non-identically distributed data sources.

The MedMNIST dataset is compiled from multiple sources, including The Cancer Imaging Archive (TCIA), the RSNA Bone Age Challenge, and the NIH Chest X-ray dataset. It is made publicly available by Dr. Bradley J. Erickson (M.D., Ph.D., Department of Radiology, Mayo Clinic) under the [Creative Commons CC BY-SA 4.0](#) license. MedMNIST offers lightweight ( $28 \times 28$ ,  $64 \times 64$ ,  $128 \times 128$ ,  $224 \times 224$ ), standardized biomedical image datasets that are well-suited for rapid experimentation and educational purposes. Biomedical data is often sensitive, distributed across multiple institutions, and subject to strict privacy regulations. By enabling collaborative model training without the need to share raw data, FL ensures compliance with privacy mandates while leveraging distributed datasets for robust and generalized model development.

During this session, you will be guided through the complete pipeline of setting up, training, and evaluating a federated learning system using the **FLOWER** framework. You will simulate multiple clients, each representing independent hospitals or laboratories, distribute the MedMNIST dataset among them, and collaboratively train your model.

## Objectives

This practical session aims to provide hands-on experience with federated learning in a biomedical context using real-world-like data. Upon completion of this assignment, you will be able:

- Understand unique challenges and requirements of federated learning in healthcare;
- Gain experience using the FLOWER framework to implement and simulate federated training across multiple clients;

- Implement federated learning solutions for medical image analysis;
- Investigate the impact of image resolution, data partitioning strategies (IID vs. non-IID), model architecture (e.g., simple CNN vs. ResNet18), and the choice of FLOWER strategy (e.g., FedAvg, FedProx) on training convergence and overall model performance.

## Computational Resources

To complete this lab assignment, you may use any of the following computational environments:

- **Google Colab** – a free cloud-based platform with limited access to GPU/TPU resources. Ideal for prototyping and small-scale experiments.
- **Kaggle Notebooks** – a cloud-based platform offering free but limited access to GPU/TPU resources.
- **Local Machine** – your own computer, if it meets the required hardware (e.g., GPU) and software setup for deep learning experiments.

## Lab variants

Each student is assigned one variant from the table below. The variant determines the dataset, applicable image resolutions, and the primary experimental focus of the assignment.

| #  | Dataset        | Resolution              | Suggested Experiments                                                                                                                                                                                                                      |
|----|----------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | PathMNIST      | 28×28 / 64×64           | Compare simple CNN vs. ResNet18 on IID/non-IID splits with FedAvg and FedProx.                                                                                                                                                             |
| 2  | ChestMNIST     | 28×28 / 128×128         | Explore binary vs. multi-label setup; test robust aggregation methods.                                                                                                                                                                     |
| 3  | DermaMNIST     | 28×28 / 64×64           | Examine effect of non-IID partitions on skin disease classification.                                                                                                                                                                       |
| 4  | OCTMNIST       | 28×28 / 64×64           | Analyze training convergence for different image resolutions and models.                                                                                                                                                                   |
| 5  | PneumoniaMNIST | 28×28 / 64×64           | Test impact of client count on performance with FedAvg.                                                                                                                                                                                    |
| 6  | RetinaMNIST    | 28×28 / 64×64 / 128×128 | Compare results using different Dirichlet values.                                                                                                                                                                                          |
| 7  | BreastMNIST    | 28×28 / 64×64           | Evaluate model robustness under 25% malicious clients.                                                                                                                                                                                     |
| 8  | BloodMNIST     | 28×28 / 64×64           | Use FedMedian and Krum under data heterogeneity.                                                                                                                                                                                           |
| 9  | TissueMNIST    | 28×28 / 64×64           | Measure influence of resolution on convergence and accuracy.                                                                                                                                                                               |
| 10 | OrganAMNIST    | 28×28 / 64×64           | Analyze ordinal regression impact on federated training.                                                                                                                                                                                   |
| 11 | OrganCMNIST    | 28×28 / 64×64           | Analyze the effect of client participation rate on model performance. Vary the fraction of active clients per round (e.g., 10%, 50%, 100%) and measure its impact on convergence speed and final accuracy under non-IID data distribution. |

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| #  | Dataset     | Resolution                 | Suggested Experiments                                  |
|----|-------------|----------------------------|--------------------------------------------------------|
| 12 | OrganSMNIST | 28×28 / 64×64 /<br>128×128 | Evaluate performance with different Flower strategies. |

## Experimental Requirements

Each student must conduct the following experiments as applicable to their assigned variant:

### 1. Data Partitioning

The dataset must be distributed across a minimum of three simulated clients. Two separate experimental configurations are required: one under IID conditions, where data is distributed uniformly across clients, and one under non-IID conditions, where the Dirichlet distribution is used to introduce statistical heterogeneity. The concentration parameter must be explicitly stated and justified in the report — in particular, the rationale for the chosen value should reflect how realistically it approximates real-world data imbalance across clients. Note that for datasets with a multi-label structure (such as ChestMNIST), applying a standard Dirichlet partition across all labels simultaneously is not straightforward; in such cases, students may either base the partition on a single dominant label or adopt an alternative heterogeneity strategy, provided the choice is clearly documented and justified. For binary classification datasets, where the Dirichlet distribution offers limited expressiveness, manual specification of per-client class proportions is an acceptable substitute.

### 2. Model Architecture

Two models are to be implemented: a lightweight convolutional neural network serving as the baseline, and a more expressive architecture such as ResNet18 or an equivalent. Both models must be adapted to match the input resolution of the assigned dataset. The report should briefly address any architectural modifications made and why they were necessary.

### 3. Federated Training

The training pipeline is to be constructed using the Flower framework, with clearly defined server and client components. The number of communication rounds is left to the student’s discretion, but must be sufficient to observe meaningful convergence behaviour or a clear plateau in the learning curves (a brief justification of the chosen number should be included in the report). Two aggregation strategies must be applied and compared: FedAvg as the standard baseline, and at least one alternative such as FedProx. The choice of the second strategy should be motivated by the experimental context, not selected arbitrarily.

### 4. Evaluation

Throughout training, the following metrics must be tracked at every communication round: training loss, validation loss, and classification accuracy. F1 score is additionally expected where class imbalance is present. Upon completion of training, final performance on the held-out test set must be reported. Where the experimental setup involves heterogeneous or adversarial client conditions, model robustness under those conditions should be assessed and discussed.

## Deliverables

Students are required to submit the following deliverables for evaluation:

### 1. Source Code

Well-organized and documented code implementing the federated learning experiments with the assigned MedMNIST variant. The codebase should cover data loading, model definition, federated training setup using the Flower framework, and evaluation procedures.

### 2. Report

A concise written report (4–10 pages) should describe the experimental setup and parameters used, present observations drawn from the results along with their interpretation in the context of federated learning, address the challenges encountered during implementation and the strategies employed to overcome them, and conclude with a summary of the key findings of the study.

## Grading Criteria

| Criterion                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Weight (%)  | Max Points |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------|
| <b>Code quality &amp; documentation</b> <ul style="list-style-type: none"> <li>• Readable, modular, well-structured code</li> <li>• Functions and classes are commented</li> <li>• Experiments are reproducible (fixed random seeds, pinned library versions)</li> <li>• README and/or requirements file provided</li> <li>• Partitioning parameters (e.g. Dirichlet <math>\alpha</math>, number of clients) are documented and justified</li> </ul>             | 25%         | 25         |
| <b>Depth of experimental work</b> <ul style="list-style-type: none"> <li>• Dataset partitioned among <math>\geq 3</math> clients; both IID and non-IID settings tested</li> <li>• At least one CNN baseline and one ResNet18 (or equivalent) compared</li> <li>• At least two aggregation strategies applied (e.g. FedAvg, FedProx)</li> <li>• Results logged per communication round</li> </ul>                                                                 | 25%         | 25         |
| <b>Quality of charts &amp; figures</b> <ul style="list-style-type: none"> <li>• Loss and accuracy/F1 curves plotted per round (train/validation)</li> <li>• All figures have labelled axes, a title, and a legend</li> <li>• Comparative plots show effect of the key variable (IID vs. non-IID, strategy, resolution, etc.)</li> <li>• Figures match the written discussion</li> </ul>                                                                          | 20%         | 20         |
| <b>Report clarity &amp; insight</b> <ul style="list-style-type: none"> <li>• Experimental setup described precisely: dataset, clients, models, hyperparameters, FL config</li> <li>• Results interpreted and discussed, not just listed</li> <li>• Convergence behaviour analysed across rounds and across conditions</li> <li>• Challenges encountered are described with the solutions applied</li> <li>• Conclusions follow logically from results</li> </ul> | 30%         | 30         |
| <b>Total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>100%</b> | <b>100</b> |